

Performance Testing

Date	4 July 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform (FinRelief AI)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing & API Performance Testing
Target Module / API	User Authentication, Loan Management API, Financial Analysis API, AI Settlement Recommendation API
Test Environment	Local Machine (FastAPI + React.js)
Test Date	1 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login API	10	30	Successful login within acceptable response time
2	Loan Submission API	15	30	Loan details stored successfully
3	Financial Analysis API	20	45	Financial health score generated correctly

4	AI Settlement Recommendation API	10	60	AI recommendation generated without errors
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.3 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.6 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	> 20 req/sec	27 req/sec	Pass	stable throughput
4	Error Rate	< 1%	0.2%	Pass	Very few request failures
5	CPU Utilization	< 80%	58%	Pass	Normal resource usage
6	Memory Utilization	< 80%	63%	Pass	Stable memory consumption

Step 4: Observations & Analysis

The FastAPI backend maintained stable performance under the tested workload. Authentication, loan management, and financial analysis APIs responded within the expected time limits. Google Gemini AI integration introduced slight latency compared to other APIs, but overall response times remained acceptable. No critical failures were observed, and system resource utilization stayed within safe limits, indicating that the application is suitable for moderate user traffic.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):